

Polynomial Regression Calculation Master Guide (Telugu Transliteration)

Ee document lo, "PolynomialRegressionCalculationNotes.pdf" lo unna step-by-step calculations ni base chesukuni, Polynomial Regression assalu ela pani chestundi, adi nijamga separate algorithm aa leda Multiple Linear Regression eena ani pin-to-pin Telugu transliteration lo (English letters tho) chala detailed ga explain chestanu.

1. The Core Secret: Polynomial Regression is NOT a Separate Algorithm!

Chala mandi students anukuntaru — Polynomial Regression ante oka kotha algorithm ani. Kani adi **ABSOLUTELY FALSE!**

Polynomial Regression ante veredo edo kadu, adi **Polynomial Feature Transformation + Multiple Linear Regression** anthe! Manam original features ni transform chesi, tarwata normal Multiple Linear Regression formula ne apply chestam.

2. Step-by-Step Calculation Walkthrough (PDF Dataset)

PDF lo ichchina exact dataset tho step-by-step calculations chuddam:

Step 1: Original Dataset

Mana daggara oka non-linear dataset undi:

X (Input)	Y (Target)
1	3
2	5
3	9
4	15
5	23

Ee data points ni graph lo plot chesthe oka **curve** la veltundi. Oka straight line ($Y = mX + b$) deenini fit cheyaledu.

Step 2: Linear Regression Equation (Standard)

Normal Linear Regression lo single feature X untundi: $Y = \beta_0 + \beta_1 X$

Step 3: Polynomial Feature Transformation (Degree = 2)

Manam degree 2 ni choose chesukuni kotha feature create chestam — adi X^2 .

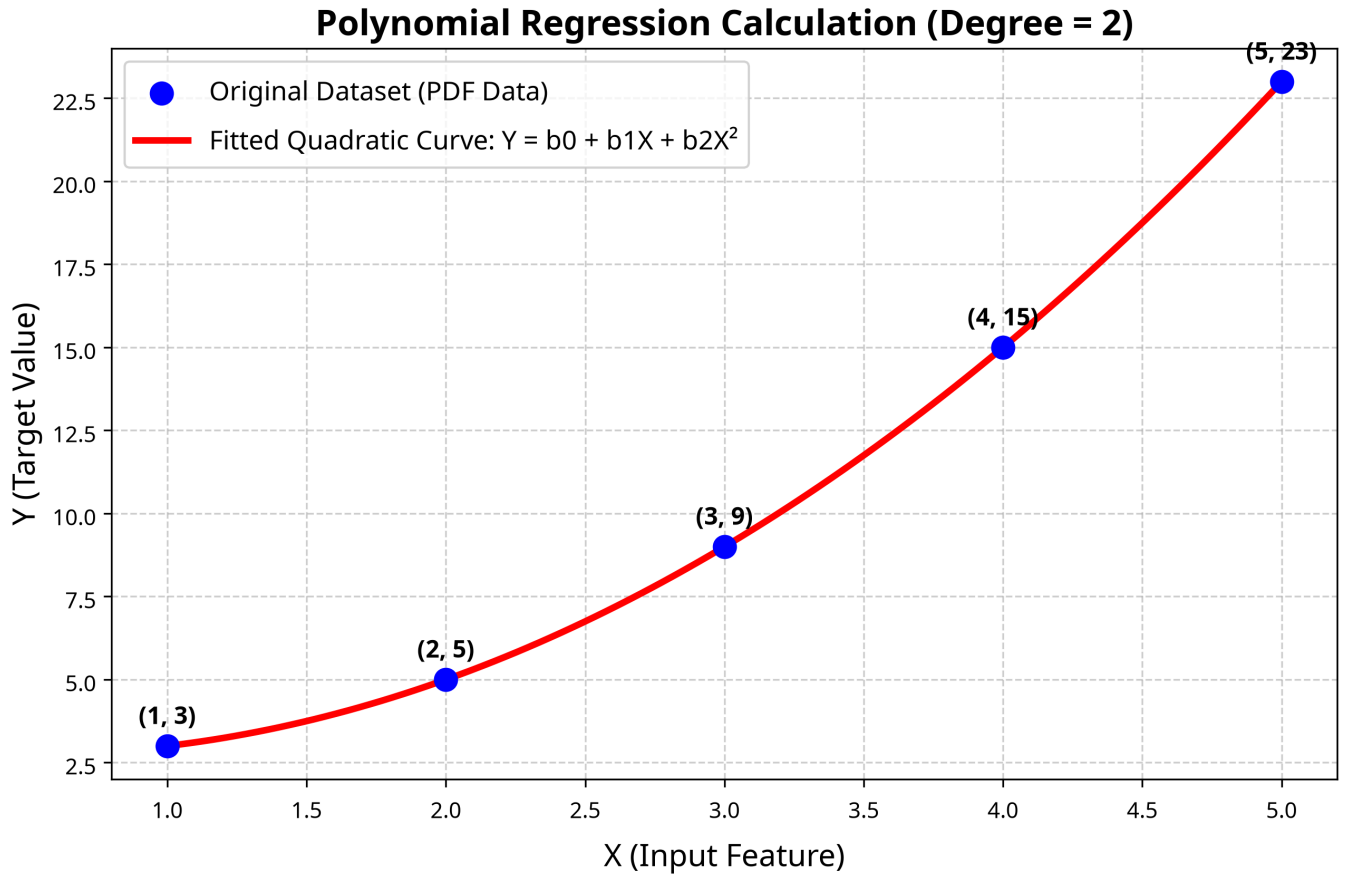
X	X^2
1	1
2	4
3	9
4	16
5	25

Ippudu dataset lo intercept (1 column), X , mariyu X^2 unnayi:

Intercept	X	X^2	Y
1	1	1	3
1	2	4	5
1	3	9	9
1	4	16	15
1	5	25	23

Visualization: Exact PDF Dataset Curve Fit

Ikkada PDF lo unna dataset ki quadratic curve ela fit avtundo chudandi:



Step 4: Rename the Features

Confusion lekunda undedaniki features ni rename chestam:

- $Z_1 = X$
- $Z_2 = X^2$

Ippudu table chudandani, adi exact ga Multiple Linear Regression dataset laage untundi:

Intercept	$Z_1 (X)$	$Z_2 (X^2)$	Y
1	1	1	3
1	2	4	5
1	3	9	9
1	4	16	15
1	5	25	23

Step 5 & 7: Multiple Linear Regression Formula

Standard Multiple Linear Regression formula: $\mathbf{Y} = \beta_0 + \beta_1 \mathbf{Z}_1 + \beta_2 \mathbf{Z}_2$

Ikkada $\mathbf{Z}_1$ badulu X mariyu $\mathbf{Z}_2$ badulu X^2 substitute chesthe: $\mathbf{Y} = \beta_0 + \beta_1 \mathbf{X} + \beta_2 \mathbf{X}^2$

Idi exact ga **Polynomial Regression equation!** Matrix form lo kuda normal equation $\beta = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{Y}$ ne use chestam.

3. Why is it Called “Linear” Regression Then? (Asalu Adi Linear Enduku?)

Mee mind lo oka question ravachu: “Idi curve kada, mari denni ‘Linear’ Regression enduku antaru?”

Answer: Machine learning lo “**Linear**” ane padam variables (X) power meeda adharapadi undadu. Adi **coefficients** (β) power meeda adharapadi untundi!

Look at the equation: $\mathbf{Y} = \beta_0 + \beta_1 \mathbf{X} + \beta_2 \mathbf{X}^2$

Ikkada coefficients ($\beta_0, \beta_1, \beta_2$) anni **power 1** lone unnayi, mariyu evaru okariki multiply avvatledu. Anduke coefficients perspective lo idi **Linear Model** eena!

4. Complete Python Implementation Code

PDF lo unna exact calculation logic ki corresponding Python code ikkada undi:

```

# =====
# Polynomial Regression Step-by-Step from PDF
# =====

import numpy as np
import pandas as pd
from sklearn.preprocessing import PolynomialFeatures
from sklearn.linear_model import LinearRegression

# Step 1: Original Dataset from PDF
X = np.array([1, 2, 3, 4, 5]).reshape(-1, 1)
Y = np.array([3, 5, 9, 15, 23])

# Step 3: Polynomial Feature Transformation (Degree = 2)
poly = PolynomialFeatures(degree=2)
X_poly = poly.fit_transform(X)

print("--- TRANSFORMED FEATURE MATRIX (Z1, Z2) ---")
print(X_poly) # Columns: [Intercept (1), X, X^2]

# Step 5 & 7: Apply Multiple Linear Regression
model = LinearRegression()
model.fit(X_poly, Y)

# Learned Coefficients
print("\n--- LEARNED COEFFICIENTS ---")
print(f"Intercept (beta_0): {model.intercept_: .4f}")
print(f"Coefficients (beta_1, beta_2): {model.coef_}")

# Predictions
Y_pred = model.predict(X_poly)
print("\n--- PREDICTIONS ---")
for actual, pred in zip(Y, Y_pred):
    print(f"Actual: {actual} | Predicted: {pred: .2f}")

```

References:

1. [PolynomialRegressionCalculationNotes.pdf](#) - Original PDF document for Polynomial Regression step-by-step calculations and matrix transformation.